

Udeshya Kumar

A learner with a strong interest in full-stack development, software engineering, and system design

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WORK EXPERIENCE

Bain Capabilities Network

Jan 2026 – Jul 2026

Data Analyst Intern

- Developed React components integrated with OpenAI APIs to generate dynamic SQL queries, execute them, analyze results, and generate user-oriented reports.
- Designed a platform-agnostic pipeline controlled through Airflow and capable of executing workloads across Snowflake and Starburst, helping reduce compute costs.
- Built and maintained Airflow DAGs to manage and schedule data pipelines from AWS S3.
- Developed Snowflake stored procedures for automated data ingestion using SQL, JavaScript, and Python.
- Imported 10+ Snowflake-dependent stored procedures into platform-agnostic Python modules.
- Created Apache Iceberg tables to enable unified data access across both Snowflake and Starburst platforms.

EmergX AI

June 2025 – July 2025

Full Stack Intern

- Contributed to the end-to-end development of the company's website.
- Created new pages and implemented dynamic routes in Next.js.
- Enhanced frontend data fetching and API interactions in React to improve functionality and responsiveness.
- Managed the database on MongoDB and improved server-side logic and state management.
- Improved application responsiveness and reduced loading times by 30% identifying and fixing memory leaks.

PROJECTS

Ultimate Tic-Tac-Toe

- A full-stack web game of Ultimate Tic-Tac-Toe (a board game composed of nine tic-tac-toe boards arranged in a 3×3 grid).
- Supports local two-player and online multiplayer modes, with shareable room invite links.
- Server-validated moves prevent cheating and enforce valid moves using predefined algorithms.
- Built with React, TypeScript, and Vite on the frontend, and Node.js, Express, and Socket.IO on the backend.

Food Scanner

- A mobile application enabling users to scan barcodes of packaged food items and retrieve nutritional values, ingredients, brand information, and more.
- Based on nutritional values and calorie count, a grade is assigned ranging from E (very bad for health) to A (good for the body in limited amounts).
- Integrated the Open Food Facts API for fetching product data.
- Built with Dart and Flutter, SQLite for local database storage.

Life-Expectancy Estimator

- Survival-analysis app combining SSA mortality tables with a Bayesian Weibull hazard model (PyMC), validated against a Cox proportional-hazards model (lifelines), trained on real NHANES health-survey data linked to mortality records.
- Takes user inputs (age, gender, BMI, blood pressure, smoking, alcohol use) to predict individualized survival probability and expected lifespan.
- Built the data pipeline in Python/pandas with an interactive Streamlit + Plotly app for posterior survival curves.

SKILLS

Programming Languages: JavaScript, TypeScript, Python, SQL, GO, Java, HTML, CSS

Frameworks, Tools & Platforms: React, Next.js, Node.js, Express, LangChain, RESTful APIs, FastAPI, Docker, Flask, Git, Bash, Android Studio, AWS, Cursor, Codex

Concepts: Data Structures and Algorithms, Object-Oriented Programming, System Design, WebSockets, CD/CI, Operating Systems, Agile Methodology, RAG, Relational and Non-Relational Databases, Encryption & Decryption

Languages: English, Hindi.

EDUCATION

Manipal Institute of Technology, Manipal

2022 – 2026

B.Tech in Computer and Communication Engineering — CGPA: 6.95

- Minor specialization in Full-Stack Development and Big Data